Duke | MARGOLIS CENTER *for* Health Policy

FUTURE OF PRIVACY FORUM

Explainable Machine Learning (XML)

What Research Review Boards Need to Know

NOVEMBER 17, 2021
1:45 PM – 2:45 PM ET

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Overview of Goals of this Presentation

- **Overview**
  - Sophisticated uses of machine learning can be a challenge for research review boards to evaluate. Savvy researchers know to use XML or explainable artificial intelligence (XAI) tools and terminology, but how do research reviewers know whether an XML or XAI tool accomplished the regulatory and ethical objectives of explanation, such as for risk-benefit assessment or informed consent forms?
- **Learning Objectives:**
  - Provide an overview of XML and XAI, and how these differ from Machine Intelligence and Artificial Intelligence
  - Identify when XML tools should be included in research proposals
  - Pair proposed machine learning methods with appropriate XML metrics and tools
  - Respond to gaps in machine learning-based research proposals and request appropriate amendments

Glossary of Acronyms

- AI (Artificial Intelligence)
- ML (Machine Learning)
- XAI (eXplainable Artificial Intelligence)
- XML (eXplainable Machine Learning)

What is– AI?

- Artificial intelligence is the computerized ability to *perform tasks* commonly associated with human intelligence, including *reasoning*, *discovering patterns* and *inferring meaning*, *generalizing*, *applying knowledge* across spheres of application, and *learning* from experience.
  - Rules based
  - Symbolic
  - Machine learning
  - Knowledge engineering
  - Natural language processing
  - Computer sensing

THE SPECTRUM OF ARTIFICIAL INTELLIGENCE

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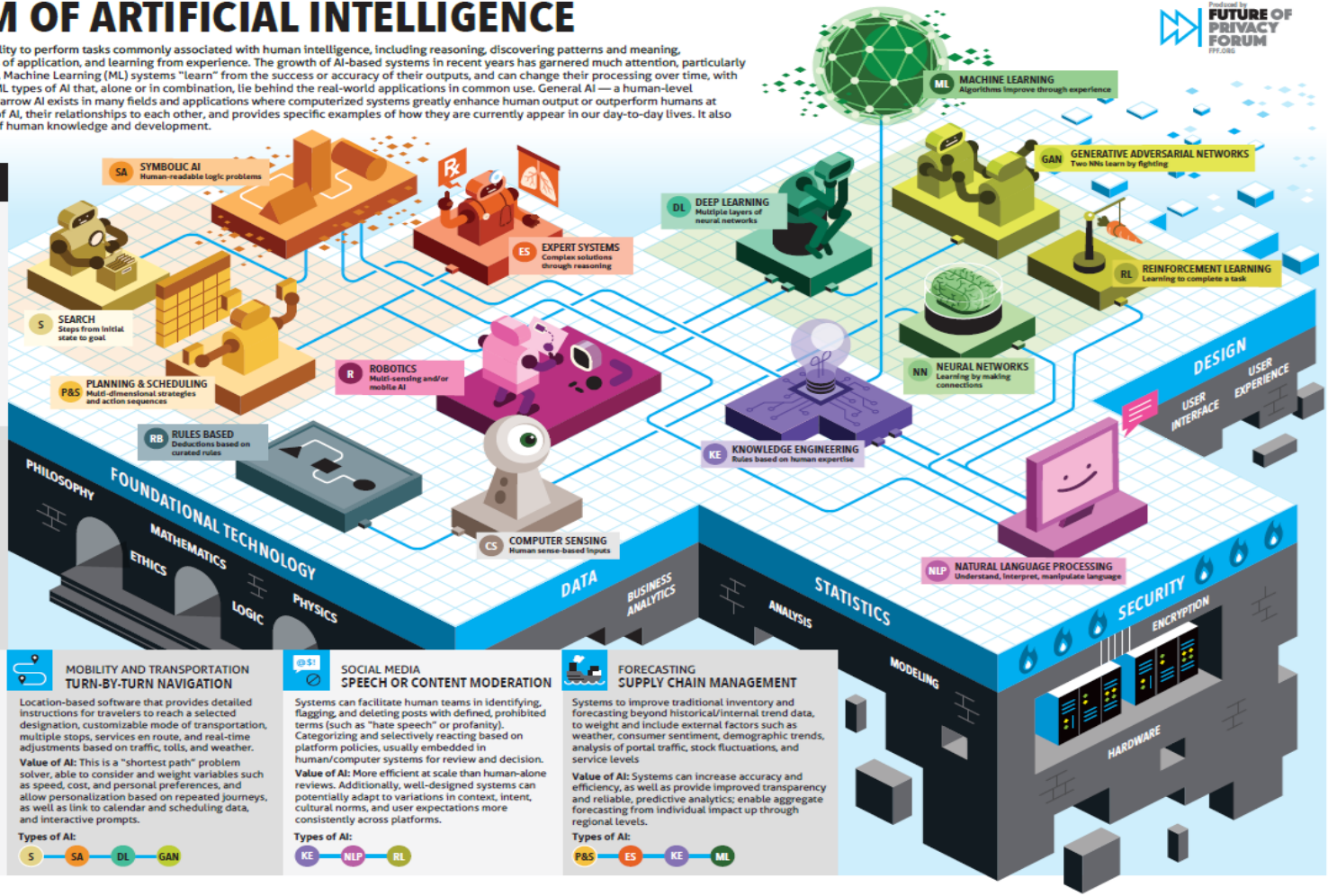
Artificial Intelligence (AI) is the computerized ability to perform tasks commonly associated with human intelligence, including reasoning, discovering patterns and meaning, generalizing, applying knowledge across spheres of application, and learning from experience. The growth of AI-based systems in recent years has garnered much attention, particularly in the sphere of Machine Learning. A subset of AI, Machine Learning (ML) systems "learn" from the success or accuracy of their outputs, and can change their processing over time, with minimal human intervention. But there are non-ML types of AI that, alone or in combination, lie behind the real-world applications in common use. General AI — a human-level computational system — does not yet exist. But Narrow AI exists in many fields and applications where computerized systems greatly enhance human output or outperform humans at defined tasks. This chart explains the main types of AI, their relationships to each other, and provides specific examples of how they are currently appear in our day-to-day lives. It also demonstrates how AI exists within the timeline of human knowledge and development.



AI USE CASES AND CONTEXTS

FINANCE
TAX COMPLIANCE

A software platform that distills tax laws into a program, creates a personalized decision system, and enables individuals to quickly and accurately file their taxes.

Value of AI: Tax compliance requires complete accuracy. This efficient, interactive system that provides precise and logically connected results allows taxpayers to understand, confirm, and have confidence in the outcome. KE provides transparent and clear explanations.

Types of AI: KE — NN — NLP

HEALTHCARE
AMBIENT CHARTING

The use of background voice-to-text processing during a patient/medical provider exchange to record those interactions into the patient's chart, along with extracting tasks, symptoms, and recommendations for further action as required.

Value of AI: Medical providers spend significant time documenting, with uneven outputs, as well as difficulty in correlating between providers. Ambient systems encode conversations, target key phrases, and present a summary for provider edit/acceptance.

Types of AI: SA — DL — NLP

TRACKING
WORKPLACE MONITORING

Embedded systems can monitor physical and digital traffic, data usage, device management, and some employee behaviors for efficiency and security management of time, assets, and resources.

Value of AI: Monitoring enables necessary enforcement of data security policies and protocols. Also, systems can monitor and manage time reporting and project management tools, as well as ensuring appropriate supervision, training, and support, including for remote workers

Types of AI: RB — CS — NN

MOBILITY AND TRANSPORTATION
TURN-BY-TURN NAVIGATION

Location-based software that provides detailed instructions for travelers to reach a selected designation, customizable mode of transportation, multiple stops, services en route, and real-time adjustments based on traffic, tolls, and weather.

Value of AI: This is a "shortest path" problem solver, able to consider and weight variables such as speed, cost, and personal preferences, and allow personalization based on repeated journeys, as well as link to calendar and scheduling data, and interactive prompts.

Types of AI: S — SA — DL — GAN

SOCIAL MEDIA
SPEECH OR CONTENT MODERATION

Systems can facilitate human teams in identifying, flagging, and deleting posts with defined, prohibited terms (such as "hate speech" or profanity). Categorizing and selectively reacting based on platform policies, usually embedded in human/computer systems for review and decision.

Value of AI: More efficient at scale than human-alone reviews. Additionally, well-designed systems can potentially adapt to variations in context, intent, cultural norms, and user expectations more consistently across platforms.

Types of AI: KE — NLP — RL

FORECASTING
SUPPLY CHAIN MANAGEMENT

Systems to improve traditional inventory and forecasting beyond historical/internal trend data, to weight and include external factors such as weather, consumer sentiment, demographic trends, analysis of portal traffic, stock fluctuations, and service levels

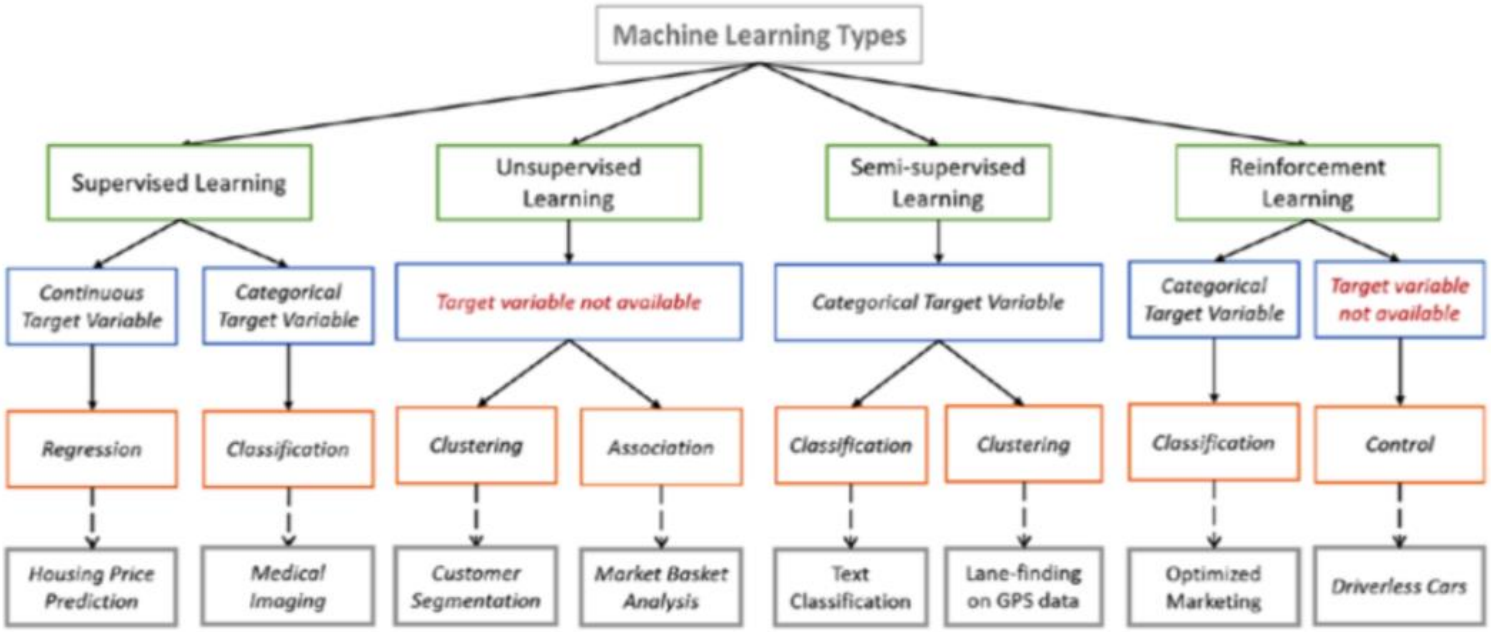
Value of AI: Systems can increase accuracy and efficiency, as well as provide improved transparency and reliable, predictive analytics; enable aggregate forecasting from individual impact up through regional levels.

Types of AI: P&S — ES — KE — ML

What is– ML?

- Computers programming that allows computers to identify patterns in data of various types (e.g., tabular, image, video, text) and to render a classification, (probability) score, cluster or other arrangement of data, or to learn how to navigate a data space.
  - Supervised
  - Semi-supervised
  - Unsupervised
  - Reinforcement

What is- ML?



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AI vs ML

Artificial Intelligence

- Encompassing definition
  - ML
  - Robotics
  - Science & Technology Studies
- “Hype cycle”

Machine Learning

- Specific algorithmic forms
- ML Performance
  - Benchmarking
  - F1 scores, AUC/ROC
- “Opacity”

What is– XAI?

- “eXplainable AI (XAI) proposes creating a suite of ML techniques that 1) produce more explainable models while maintaining a high level of learning performance (e.g., prediction accuracy), and 2) enable humans to understand, appropriately trust, and effectively manage the emerging generation of artificially intelligent partners. XAI draws as well insights from the Social Sciences and considers the psychology of explanation” (Barredo Arrieta et al 2020, p. 83)

What is- XAI?

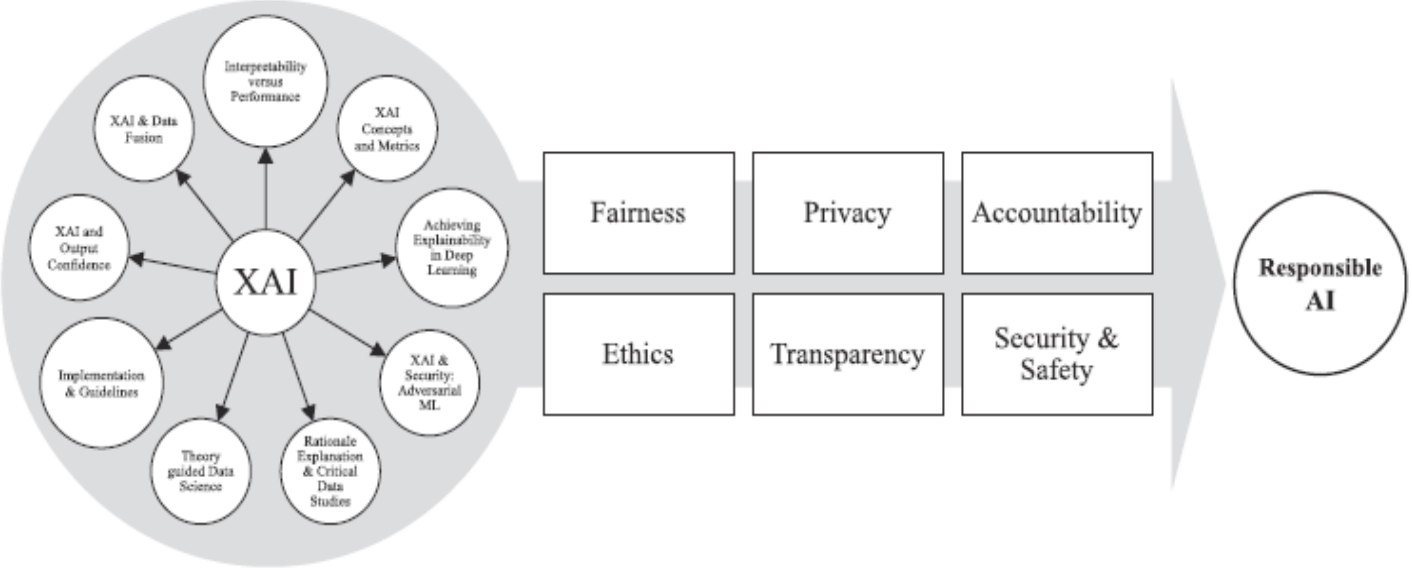

Fig. 14. Summary of XAI challenges discussed in this overview and its impact on the principles for Responsible AI.

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What is– XAI?

- “Given an audience, an explainable artificial intelligence is one that produces details or reasons to make its functioning clear or easy to understand” (Barredo Arrieta 2020, p. 85)

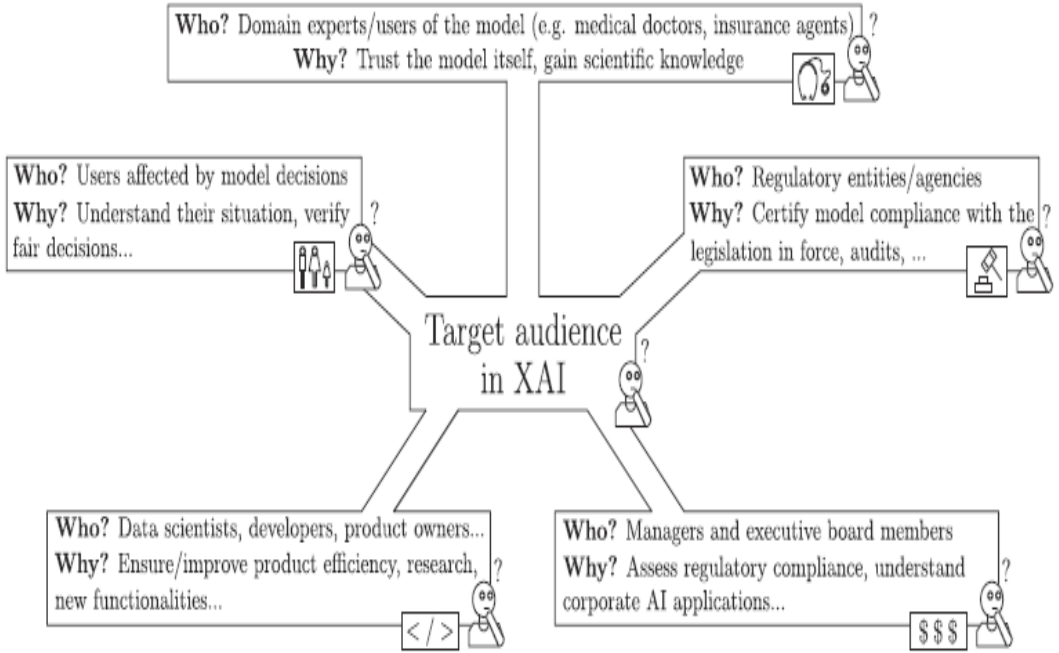

Fig. 2. Diagram showing the different purposes of explainability in ML models sought by different audience profiles. Two goals occur to prevail across them: need for model understanding, and regulatory compliance. Image partly inspired by the one presented in [29], used with permission from IBM.

What is- XML?

- Explainable Machine Learning includes techniques that allow machine learning to be explained to lay users
  - Simulatability—an algorithm can be decomposed into elements that an average human cognition can repeat
  - Decomposability- the inputs of a system and its actions can be made understandable without additional tools
  - Transparency- an algorithm that is explorable by means of mathematical analysis
- Examples- text explanations, visual explanations, local explanations, explanations by example, explanations by simplification, and feature relevance explanation

XAI vs XML

- Explanations drawn with respect to the audience
- NIST's new guidance on XAI
  - Sufficient explanations allow audiences to form an actionable mental model of how an AI works

- Explanations drawn with respect to the algorithmic form
- New bias-monitoring technologies
  - Amazon's clarify

Tools for Explainability

Triggers	User/Learner's Goal	
	Need to Understand	Need to Accomplish
How do I use it?	Use	Primary task goals
How does it work?	Mechanism, Understandability	Feeling of Satisfaction
What did it just do?	Mechanism, Understandability	Feeling of Sufficiency
What does it achieve?	Function	Usefulness
What will it do next?	Observability	Trust, Reliance
How much effort will this take?	Usability	Primary Task Goals
What do I do if it gets it wrong?	Anticipation	Control
How do I avoid the failure modes?	Surprise	Adaptation, Work-around
What would it have done if x was different?	Mechanism, Understandability	Curiosity, Trust, Reliance
Why didn't it do z?	Anticipation, Surprise	Curiosity, Trust, Reliance

Mueller, S. T., Hoffman, R. R., Clancey, W., Emrey, A., & Klein, G. (2019). Explanation in human-AI systems: A literature meta-review, synopsis of key ideas and publications, and bibliography for explainable A, p. 90.

Tools for Explainability

Format (How it is expressed)	Reference (What it is about)
• Visualizations (e.g., Heat maps) • Text (Statements, Narratives or Stories, Answers to queries, Human-machine dialogs) • Formal Expressions (Logical expressions, Matrices) • Conceptual Process Models (Diagrams) • Graphs, Networks • Tables • Abstractions, generalizations • Timelines • Hierarchies (Trees)	• Examples (includes misclassifications, counter-examples, outliers, clear cases, close competitors) • Patterns, Classes, Ontologies • Features, Weights, Probabilities, Ranks, Parameters • Decisions, Strategies, Goals • Algorithms, Computational Processes, Proofs • Incidents, Events (includes self-explanations or stories) • Cause-effect relations

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Fairness metrics

Fairness metrics	Bias preserving?
1. Group fairness, Statistical (demographic) parity[i]	✗
2. Conditional statistical (demographic) parity, Conditional independence[ii]	✗
3. Predictive parity, outcome test[iii]	✓
4. False positive error rate balance[iv]	✓
5. False negative error rate balance,[v] Equal opportunity[vi]	✓
6. Equalized odds[vii]	✓
7. Conditional use accuracy equality[viii]	✓
8. Overall accuracy equality[ix]	✓
9. Treatment equality[x]	✓
10. Test-fairness or calibration[xi]	✓
11. Well-calibration[xii]	✓
12. Balance for positive class[xiii]	✓
13. Balance for negative class[xiv]	✓
14. Causal discrimination[xv] (direct discrimination)	*
15. Fairness through unawareness[xvi]	*
16. Fairness through awareness[xvii]	✗
17. Counterfactual fairness[xviii]	✗
18. No unresolved discrimination[xix] (causal variant of 2)	✗
19. No proxy discrimination[xx]	✗
20. Path based causal reasoning[xxi]	✗

Wachter S., Mittelstat B. & Rusell C. (2021). Bias Preservation in Machine Learning: The Legality of Fairness Metrics Under EU Non-Discrimination Law. In: *West Virginia Law Review, Forthcoming.*

AI and data protection risk mitigation and management

- Published by the Information Commissioner's Office (ICO).
- The toolkit is designed to help risk professionals identify and mitigate the data protection risks that AI systems cause or exacerbate.
- 13 perspectives, including
  - transparency,
  - statistical accuracy,
  - discrimination and
  - data minimization.

URL: https://ico.org.uk/about-the-ico/ico-and-stakeholder-consultations/ai-and-data-protection-risk-mitigation-and-management-toolkit/.

Number	Risk	How AI can create or exacerbate this risk	Probability (1 = Low, 2 = low/medium, 3 =	Severity (1= low, 2 = low/medium, 3 =	Risk score (probability x severity)	Current status of the risk	Practical steps you could take	Intended action(s)
10,1	Using AI to process personal data without informing individuals impinges on human dignity and a breach of UK GDPR Articles 13, 14 and 22.	AI systems may be opaque, undermining transparency and interpretability of decisions.			0		o Publish privacy information that is concise, transparent, intelligible and uses clear and plain language. o Ensure the privacy information includes meaningful information about the logic involved, as well as the significance and envisaged consequences of the processing for individuals. o Tailor explanations to the intended audience so that they are clear and easy for individuals to understand, considering the level of knowledge that the explanation recipient has about the subject. o Ensure that the privacy information includes the purposes of the processing and the lawful bases (and the legitimate interests for the processing if applicable). o Explain any trade-offs to individuals or any human tasked with reviewing AI output. o Provide privacy information to individuals at the time their personal data is collected (unless an exemption applies). o Ensure individuals are made aware of the privacy information and have an easy way to access it, eg using a combination of appropriate techniques, such as a layered approach, dashboards, just-in-time notices, icons and mobile and smart device functionalities. o Provide details enabling individuals to challenge the outcome if they think it is flawed (eg if some of the input data was incorrect or irrelevant, or additional data wasn't taken into account that the individual thinks is relevant).	
10,2	Failure to provide information to individuals where that information was not obtained from the individual leads to unfair invisible processing, a reduction of human autonomy and a	Datasets containing personal data that are used for training AI purposes may be shared among multiple organisations without the individual's knowledge.					o Carry out due diligence of potential data suppliers to confirm appropriate privacy information has been provided to data subjects. o Carry out a DPIA to determine whether providing privacy information would involve a disproportionate effort when	

... | 5. Lawful Basis | 6. Trade offs | 7. Statistical accuracy | 8. Discrimination | 9. Security & integrity | 10. Transparency | 11. Data minimisation | 12. Individual rights | 13. Human review

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Number	Risk	How AI can create or exacerbate this risk	Probability (1 = Low, 2 = ow/medium, 3 =	Severity (1= low, 2 = ow/medium, 3 =	Risk score (probability x severity)	Current status of the risk	Practical steps you could take	Intended action(s)
7,1	A lack of structured testing processes in place leads to pre-implementation testing not being undertaken or completed effectively.	There are several different ways of measuring the statistical accuracy of an AI system.			0		o Put in place a policy / documented process that includes details of how the system will be tested prior to implementation. o Ensure the 'test plan' includes all the relevant checks to ensure that there are no errors in data outputs or statistical errors. o Ensure the test plan includes documented tolerances for errors. o Document the results of the testing. o Ensure that acceptance of the test results is signed off by management. o Retrain the AI system following testing (eg by improving input data, different balance of false positives and negatives, or using different learning algorithms). o Test the AI system on new data set(s) to confirm the same outcome is reached. o Ensure that testing was completed prior to the deployment of the AI system.	
7,2	Failure to have structured testing processes in place leads to human review not being undertaken or completed effectively.	Complex AI systems may lead to automation bias.			0		o Put in place a policy / documented process that includes details of the methodology that will be used by a human reviewer when testing the system for statistical accuracy. o Put in place a testing process that outlines the test plan criteria, requirements, and sampling method / size. o Ensure the 'test plan' includes all the relevant checks to ensure that the rate of error in data outputs or statistical errors is within acceptable and documented tolerances. o Ensure the test plan includes documented tolerances for errors. o Document the results of the testing. o Ensure that acceptance of the test results is signed off by management.	

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8,1	Failure to review the system's design leads to bias or discriminatory outputs being built into the system	AI systems may lead to unjust discriminatory outcomes if training data is discriminatory.			0		o Ensure that risks are drawn from policy, user research and design, and computer science expertise. o Ensure that the review includes all the relevant checks to the design of the AI systems to ensure that there isn't the potential for discriminatory outputs or decisions to be made. o Ensure that reviews are conducted by appropriately skilled and experienced technical 'experts'. o Document the results of the reviews, including any trade-offs and their technical implications o Ensure that reviews are signed off by senior management. o Document where the AI system design has been adapted / changed following the review where necessary.	
8,2	Failure to use protected characteristics to ensure fairness/accuracy (as defined by the goals of the system) leads to data subjects facing inaccurate or unfair results.	AI systems may use other data points as a proxy for characteristics leading to unlawful discriminatory outcomes that are difficult to detect.			0		o Document considerations and the results of any internal consultations. o Document where possible positive discriminatory characteristics have been included in the system. o Ensure these characteristics are tested thoroughly to ensure they produce the right outputs consistently.	
8,3	No structured testing process in place leads to pre-implementation testing not being undertaken or completed effectively.	There are several different ways of measuring the statistical accuracy of an AI system.			0		o Put in place a policy / documented process that includes details of how the system will be tested prior to implementation. o Ensure the 'test plan' includes all the relevant checks to ensure that there are no discriminatory outputs or decisions (or that these decisions / outputs are at a necessary minimum level) made within AI systems. o Document the results of the testing. o Ensure that the acceptance of the test results is signed off by senior management. o Document where the AI system has been 'retrained' following testing (eg by implementing algorithmic fairness measures / fairness constraints).	

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11,1	A lack of reviews at each stage of the AI lifecycle leads to a risk of inappropriate retention of data and non-compliance with Article 5 (1) (c)	AI systems generally require large amounts of data			0		o Assess whether the features used to train the AI system – and therefore the data – are relevant for the purpose, and make sure in the design stage only that data is processed o Do an assessment to ensure the training data can be modified to reduce the extent to which it can be traced back to specific individuals o Include specific review phases in your development plan to check data is being minimised / not retained when no longer needed. o Do an assessment that includes a justification for the retention of data where applicable o Remove or delete irrelevant data during the system development phase. Maintain a record of what data was removed or deleted.
11,2	Data that is not assessed for its relevance and then separated and deleted leads to excessive data being retained for longer than is necessary and non-compliance with Article 5 (1), (c).	AI systems may find some data points statistically irrelevant.			0		o Map out all the processes in which personal data is used in the different phases of an AI system. o Ensure that the mapping and then subsequent assessment for the potential minimisation of data includes data used in the production of the system and then as part of ongoing research to retrain the system o Index the personal data used in each phase of the AI system lifecycle. o Design a process that detects any duplicated data present in different phases (from production to research) and delete where necessary.

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FAT/ML

- Five Principles for Accountable Algorithms and a Social Impact Statement for Algorithms

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

FAT/ML

- Five Principles for Accountable Algorithms and a Social Impact Statement for Algorithms

Explainability

Ensure that algorithmic decisions as well as any data driving those decisions can be explained to end-users and other stakeholders in non-technical terms.

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

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Ensure that algorithm
other stakeholders in

Explainability

Guiding Questions

- Who are your end-users and stakeholders?
- How much of your system / algorithm can you explain to your users and stakeholders?
- How much of the data sources can you disclose?

Initial Steps to Take

- Have a plan for how decisions will be explained to users and subjects of those decisions. In some cases it may be appropriate to develop an automated explanation for each decision.
- Allow data subjects visibility into the data you store about them and access to a process in order to change it.
- If you are using a machine-learning model:
  - consider whether a directly interpretable or explainable model can be used.
  - describe the training data including how, when, and why it was collected and sampled.
  - describe how and when test data about an individual that is used to make a decision is collected or inferred.
- Disclose the sources of any data used and as much as possible about the specific attributes of the data. Explain how the data was cleaned or otherwise transformed.

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Fairness

Ensure that algorithmic decisions do not create discriminatory or unjust impacts when comparing across different demographics (e.g. race, sex, etc).

We have left some of the terms above purposefully under-specified to allow these principles to be broadly applicable. Applying these principles well should include understanding them within a specific context. We also suggest that these issues be revisited and discussed throughout the design, implementation, and release phases of development. Two important principles for consideration were purposefully left off of this list as they are well-covered elsewhere: privacy and the impact of human experimentation. We encourage you to incorporate those issues into your overall assessment of algorithmic accountability as well.

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

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Fairness

Guiding Questions

- Are there particular groups which may be advantaged or disadvantaged, in the context in which you are deploying, by the algorithm / system you are building?
- What is the potential damaging effect of uncertainty / errors to different groups?

Initial Steps to Take

- Talk to people who are familiar with the subtle social context in which you are deploying. For example, you should consider whether the following aspects of people's identities will have impacts on their equitable access to and results from your system:
- Race
- Sex
- Gender identity
- Ability status
- Socio-economic status
- Education level
- Religion
- Country of origin
- If you are building an automated decision-making tool, you should deploy a fairness-aware data mining algorithm. (See, e.g., the resources gathered at http://fatml.org).
- Calculate the error rates and types (e.g., false positives vs. false negatives) for different sub-populations and assess the potential differential impacts.

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Fairness

Responsibility

Make available externally visible avenues of redress for adverse individual or societal effects of an algorithmic decision system, and designate an internal role for the person who is responsible for the timely remedy of such issues.

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Explainability

Fairness

Responsibility

Guiding Questions

- Who is responsible if users are harmed by this product?
- What will the reporting process and process for recourse be?
- Who will have the power to decide on necessary changes to the algorithmic system during design stage, pre-launch, and post-launch?

Initial Steps to Take

- Determine and designate a person who will be responsible for the social impact of the algorithm.
- Make contact information available so that if there are issues it's clear to users how to proceed
- Develop a plan for what to do if the project has unintended consequences. This may be part of a maintenance plan and should involve post-launch monitoring plans.
- Develop a sunset plan for the system to manage algorithm or data risks after the product is no longer in active development.

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Accuracy

Identify, log, and articulate sources of error and uncertainty throughout the algorithm and its data sources so that expected and worst case implications can be understood and inform mitigation procedures.

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Explainability

Fairness

Responsibility

Accuracy

Guiding Questions

- What sources of error do you have and how will you mitigate their effect?
- How confident are the decisions output by your algorithmic system?
- What are realistic worst case scenarios in terms of how errors might impact society, individuals, and stakeholders?
- Have you evaluated the provenance and veracity of data and considered alternative data sources?

Initial Steps to Take

- Assess the potential for errors in your system and the resulting potential for harm to users.
- Undertake a sensitivity analysis to assess how uncertainty in the output of the algorithm relates to uncertainty in the inputs.
- Develop a process by which people can correct errors in input data, training data, or in output decisions.
- Perform a validity check by randomly sampling a portion of your data (e.g., input and/or training data) and manually checking its correctness. This check should be performed early in your development process before derived information is used. Report the overall data error rate on this random sample publicly.
- Determine how to communicate the uncertainty / margin of error for each decision.

algorithm. (See, e.g., the resources gathered at http://fatml.org).

- Calculate the error rates and types (e.g., false positives vs. false negatives) for different sub-populations and assess the potential differential impacts.

Explainabili

Fairness

Responsibilit

Accuracy

Identify, log, and articula
expected and worst case

covered elsewhere: priv
issues into your overall

URL: https://www.fatml.org/

FAT/ML

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Explainability

Fairness

Responsibility

Accuracy

Guiding Questions

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Auditability

Enable interested third parties to probe, understand, and review the behavior of the algorithm through disclosure of information that enables monitoring, checking, or criticism, including through provision of detailed documentation, technically suitable APIs, and permissive terms of use.

algorithm. (See, e.g., the resources gathered at http://fatml.org).

- Calculate the error rates and types (e.g., false positives vs. false negatives) for different sub-populations and assess the potential differential impacts.

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Explainability

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Responsibility

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Auditability

Guiding Questions

- Can you provide for public auditing (i.e. probing, understanding, reviewing of system behavior) or is there sensitive information that would necessitate auditing by a designated 3rd party?
- How will you facilitate public or third-party auditing without opening the system to unwarranted manipulation?

Initial Steps to Take

- Document and make available an API that allows third parties to query the algorithmic system and assess its response.
- Make sure that if data is needed to properly audit your algorithm, such as in the case of a machine-learning algorithm, that sample (e.g., training) data is made available.
- Make sure your terms of service allow the research community to perform automated public audits.
- Have a plan for communication with outside parties that may be interested in auditing your algorithm, such as the research and development community.

algorithm. (See, e.g., the resources gathered at http://fatml.org).

- Calculate the error rates and types (e.g., false positives vs. false negatives) for different sub-populations and assess the potential differential impacts.

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XML and XAI for Health Data Applications

- Slido poll: https://app.sli.do/event/kfho05au/live/polls
- Statement #1 (agree/disagree): XAI/XML should be legally required in health-related contexts.
- Statement #2 (agree/disagree): My institution is concerned about the explainability of AI/ML in health-related contexts.

XML and XAI for Health Data Applications: Contextual Overview

- In real-world healthcare and health consumer contexts, trustworthiness in the interpretation of health data and explainable AI/ML have become moral, fairness, and trustworthiness imperatives for high performing AI/ML models.
- XML and XAI is therefore a moral imperative across high-risk and precarious healthcare or health consumer scenarios, which include but are not limited to:
  - Predict surgery risks and outcomes
  - Conduct autonomous robotic surgeries
  - Detect, predict, and/or diagnose disease
  - Conduct pathology assessments
  - Radiological assessments
  - Predicting antibiotic resistance
  - Predicting patient mortality in intensive care or emergency units
  - Triaging or prioritizing treatment of emergency unit patients
  - Hospital discharge decision support
  - Direct-to-consumer medical AI/ML applications

XML and XAI for Health Data Applications: Legal Implications and Considerations

- XAI and XML are a legal necessity (although not always legally-required) across various health-related contexts:
  - Accountability
  - Liability
  - Culpability
- *Case examples*
  - Article 22(3) of the European General Data Protection Regulation (GDPR) states that data subjects that are legitimately subjected AI/ML-generated decisions have a right “to express his or her point of view and to contest the decision.”
    - This would include a data subject’s ability to contest AI/ML-driven health-related decisions.
    - Thus, in GDPR jurisdictions and other jurisdictions with mirroring legal requirements, XAI/XML may be legally-required in health-related scenarios.
  - The United States Food and Drug Administration discussed, within their AI/ML-Based Software as a Medical Device (SaMD) Action Plan, their proposed framework that introduces the principle of a “Predetermined Change Control Plan”
    - SaMD Pre-Specifications explain "what" aspects the manufacturer intends to change through learning, and the
    - Algorithm Change Protocol (ACP) explains "how" the algorithm will learn and change while remaining safe and effective.

XML and XAI for Health Data Applications: Ethical vs. Legal Considerations

- In instances where XAI/XML are legally required, interpretation requirements may vary:
  - Maximal interpretation of XAI/XML
    - Example: Discuss why the diagnosis, procedure, or clinical conclusion was the scientifically best possible explanation following an assessment of clinical presentations, signs, symptoms and indicators.
  - Minimal interpretation of XAI/XML
    - Example: Simply state that a diagnosis, procedure, or clinical conclusion informed or guided by a machine on the basis of health data.

Maximal interpretations can be ethical, yet not legally required across varied legal jurisdictions and real-world, health-related contexts (both research and practice).

XML and XAI for Health Data Applications: Data Quality, Sourcing, and Other Important Considerations

- XAI/XML models still run the risk of being:
  - Poor Quality (e.g., inaccurate, incomplete, outdated, skewed)
    - Models built on poor quality data can lead to diagnostic inaccuracies or a diagnostic odyssey that may affect patient or health consumer choices and outcomes, mislead clinician decision making, introduce risks and liabilities, etc.
  - Sourced or procured through unethical or unauthorized means from various data sources
    - Models can be built using social determinants of health data (e.g., social media posts, geolocation history, ride-sharing history, etc.) sourced or procured in questionable ways or through means that are unbeknownst to the data subject(s)
  - Discriminatory
    - Models that are built on or fed datasets that contain 'triggers' of discrimination (e.g., gender, age, race/ethnicity, etc.) naturally introduce risk of bias depending on how the XAI/XML are trained to respond to such triggers

XML and XAI in the Research Setting

XAI

- Ethics in AI
  - Fairness and bias
  - Transparency
- User acceptance of AI
- Critical studies of AI

XML

- Interpretability
- Understandability
- Explicability
- Utility
  - Does data utility change when ethical controls are applied?

XML and XAI in a Research Protocol

XAI

- Lay or non-specialist audience focus
- UX/UI focus
- Ethics focus

XML

- Specialist audience focus
- Benchmark performance comparisons between research data and known datasets
- Specialty programs in bioinformatics, MD/PhD programs, Computational clinicians

XML and XAI in a Research Protocol

XAI

- Human subjects concerns
  - Surveys
  - Interviews
  - Observations
- Animal care and use
  - AI systems in an animal environment

XML

- Human subjects concerns
  - Students as data sources
  - Data source identifiability
- Biosafety/ biosecurity
  - Select agent genetic data

XML and XAI in a Risk-Benefit Profile

- Human computer interaction studies typically pose minimal risk to participants but interaction with physical systems driven by probabilistic guidance systems may increase risk to slightly above that which we experience in daily life
- User experience/ User interaction studies are a common part of daily digital life; in a research setting this can involve deception or use of a “standard of care” vs ‘therapeutic’ treatment
- AI’s “therapeutic misconception” and automation bias

XML and XAI in an Informed Consent Form

- Can subjects truly consent to participation in AI/ML studies?
  - Information, comprehension, voluntariness
  - NIST “Interpretable AI” discussion: can subjects form a “mental model” that adequately simulates the tasks of the AI or ML system?
- Do subjects have access to a comparable non-automated system?
- Can subjects consent to being represented in datasets for benchmarking ML performance?
- Are faculty subjecting students to untested AI/ ML systems for educational research?

Research Advertisement

- Are you advertising research opportunities via internet advertising platforms?
  - If so, AI may be part of the service of those ads
- For platforms used in research surveys and other subject interaction environments, what is the AI used?
  - Researchers need to understand the Terms of Service and EULA and be able to explain the risks of the platform

Upcoming Sessions

Beyond the Basics of Online Advertising

November 30, 2021

Learn More & Register Today at: fpf.org/training

FUTURE OF PRIVACY FORUM

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